

1 Synthetic Recovery Validation Derivation

1.1 Input and Assumption Nodes

Input node: `estimator.beta_map`. Data are generated from `model.gaussian_observation`.

1.2 Derivation

For each n , seed, and true coefficient β_0 , compute

$$e_n = \|\hat{\beta}_{MAP,n} - \beta_0\|_2.$$

Average e_n across seeds.

1.3 Result

The validation result is a decreasing mean error trend as sample size increases.